

Who Profited from the PPP? An analysis of business owners ^{*}

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Abstract

This paper examines the Paycheck Protection Program’s effect on business owners using de-identified IRS tax records linked to the universe of PPP disbursements. We document three findings. First, about \$100 billion (19 percent) of 2020 PPP was potentially ineligible—flowing to fictitious entities, ineligible businesses, or in excess of statutory limits—much of it detectable from government records available before disbursement. Second, among eligible recipients, after-tax business profits rose 134 percent while worker compensation increased just 0.7 percent; bounding exercises suggest two-thirds to three-quarters of benefits accrued to owners rather than workers. Third, over half of PPP dollars flowed to owners in the top 1 percent of the income distribution, boosting their after-tax income by 42 percent—an order of magnitude larger than the 3.4 percent increase from the Tax Cuts and Jobs Act. The exclusion of forgiven PPP from taxable income saved owners an estimated \$145 billion, further magnifying the concentration of benefits among high-income households.

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1 Introduction

The Paycheck Protection Program (PPP) was the largest business support program in U.S. history, disbursing over \$800 billion in forgivable loans to millions of firms during the COVID-19 pandemic. The economic literature has focused on the PPP’s employment effects, with studies varying widely in the estimated number of jobs saved.¹ The fact that PPP was available to nearly all firms makes it challenging to estimate employment effects of the program, especially outside of the short-run. This feature also complicates the study of incidence — namely, whether owners or workers captured the majority of the program’s gains and how those benefits were distributed across the income distribution - a topic that has received little attention. Departing from the approach and focus of the prior literature, we examine how recipient firms and business owners fared during the pandemic to make progress on these questions.

Using IRS tax records linked to the universe of 2020 PPP loans, we document three central findings. First, about \$100 billion (19 percent) of PPP was potentially ineligible—flowing to fictitious entities, ineligible businesses, or in excess of statutory limits—and much of this was detectable from government records available before disbursement. Second, among eligible recipients (who received on average \$103k of PPP), after-tax business profits rose 134 percent (\$100k), suggesting that the vast majority of benefits accrued to owners rather than workers. Third, PPP benefits were concentrated among the wealthiest business owners: over half of PPP dollars flowed to owners in the top 1 percent of the income distribution, boosting their after-tax income by 42 percent—an order of magnitude larger than the 3.4 percent annual increase from the Tax Cuts and Jobs Act.

Business owners benefited from the PPP because of several factors. First, the program covered a substantial share of firms’ payroll and other operating expenses, which directly increased profits. Second, the magnitude of PPP payments was large relative to the impact

¹These estimates, ranging from as low as 1 million to over 10 million jobs saved at its peak, rely on different identifying assumptions and types of firms affected.

of the pandemic on business revenues and profits, particularly in less-affected sectors. As we show, these factors combined to more than double the after-tax income of PPP-recipient firms in 2020 compared to 2019. Finally, forgiven PPP loans were excluded from owners' taxable income (even as wage payments continued to be deductible and taxable to employees), shielding these higher profits from tax. As a result, a significant fraction of the program's benefits accrued to business owners. Because business ownership in the United States is concentrated among higher-income individuals, PPP benefits disproportionately accrued to high-income households.

To understand the extent to which business owners benefited from PPP, this paper links PPP disbursements to businesses and business owners IRS tax records and describes the financial profiles and dynamics of the individuals and firms who applied for and received PPP funds.

The analysis includes four key steps. First, we identify how much of PPP flowed to eligible versus ineligible businesses, and verify that the amounts paid followed statutory limits. Our analysis finds that about \$100 billion of the \$529 billion in 2020 PPP disbursements (or almost 20 percent) flowed to ineligible businesses or exceeded statutory limits—including about \$31 billion to fictitious entities or businesses with no payroll, \$18 billion to businesses ineligible by size or operational status, and \$46 billion in excess of statutory amounts. Most of the information needed to flag these payments was available in government databases before funds were disbursed.

In the second step, we describe how recipient firms fared during the pandemic and how PPP affected firm profitability.² PPP substantially boosted the profits of firms and the incomes of their owners. In 2019, firms that later received PPP earned about \$244 billion in after-tax profits. In 2020, absent PPP, after-tax profits fell modestly to \$210 billion—a decline of \$33 billion. But these firms received \$417 billion in PPP funds, and because PPP was excluded from taxable income, after-tax PPP-inclusive profits rose to \$628 billion—157

²"Affected" in an accounting sense, though in a simple model, PPP should weakly decrease pre-PPP firms profits.

percent above 2019. We estimate the tax exclusion saved business owners \$145 billion in income taxes. The second round of PPP disbursed in 2021 was intended to be more targeted to firms negatively impacted by the pandemic but nevertheless resulted in similarly large increases in firm profits because firms had already recovered by the time the funding was disbursed.

The design of the PPP, its special tax treatment, and the large increase in firm profits suggest that the majority of the program’s benefits accrued to firm owners rather than workers, which is consistent with some of the literature’s estimates of PPP on employment (Autor et al., 2022a; Granja et al., 2023). We then undertake a set of back-of-the-envelope bounding exercises on the incidence that suggest the large majority of the benefits accrued to firm owners, not workers.

We also examine the distribution of PPP benefits across the income distribution of firm owners. Ranking PPP recipient owners by 2020 total income including PPP, about 56 percent of PPP funds accrued to taxpayers in the top 1 percent and 86 percent to the top 10 percent. As a result, the after-tax income of PPP-recipient business owners in the top 1 percent increased by about 42 percent in 2020 relative to 2019.

This paper makes the following contributions. First, using IRS and Treasury administrative records that were available at the time of disbursement, we identify approximately \$100 billion in preventable ineligible payments—including loans to fictitious entities, ineligible businesses, and amounts exceeding statutory limits. Second, we document the PPP windfall to firm profits and provide transparent incidence bounds—using program rules, firm-level outcomes, theory, and/or external employment estimates—showing that at least two-thirds to three-quarters of benefits accrued to owners, not workers. Third, we allocate PPP to individual owners and show extreme concentration at the top of the income distribution, with the \$145 billion tax windfall from PPP’s exclusion from income magnifying this regressivity.

The remainder of the paper proceeds as follows. Section 2 provides background on PPP’s legislative history, program rules, and the existing literature on its employment effects.

Section 3 describes our data linkage between SBA loan records and IRS tax filings. Section 4 examines program integrity and the mechanical effects of PPP on firm profits. Section 5 presents bounding exercises for owner versus worker incidence. Section 6 allocates PPP to owners and documents the distribution of benefits across the income distribution. Section 7 concludes with implications for program design and tax policy.

2 Background

In 2020, the Coronavirus Aid, Relief, and Economic Security (CARES) Act introduced the Paycheck Protection Program (PPP), which ultimately disbursed about \$525 billion in forgivable loans in 2020 to small businesses, certain nonprofits, tribal businesses, and veterans' organizations that were operational on or before February 15, 2020. Generally, any entity with 500 or fewer employees—or that otherwise met the Small Business Administration's size standards—qualified, including sole proprietors and self-employed individuals. The maximum loan amount equaled 2.5 times a borrower's average monthly payroll costs (capped at \$10 million), and borrowers were required to maintain payroll to qualify for loan forgiveness. The PPP Flexibility Act of June 2020 significantly relaxed forgiveness requirements, reducing the effective payroll maintenance threshold to roughly 27 percent of prior payroll (see Appendix C for details). As a result, the PPP's incentive to retain workers at pre-pandemic levels was strongest in April and May 2020, before these rules were relaxed.

The CARES Act specified that forgiven PPP loans would not be included in income for tax purposes. In April 2020, the IRS issued Notice 2020-32 (Internal Revenue Service, 2020), which stated that expenses paid with forgiven PPP funds are not deductible, following a long-standing principle (IRC §265) that no deduction is allowed for expenses allocable to wholly tax-exempt income, to prevent taxpayers from obtaining a “double benefit.” However, in December 2020—long after PPP funds were disbursed—the Consolidated Appropriations Act, 2021 (United States Congress, 2020) reversed this position, allowing full deductibility

of expenses paid with forgiven PPP proceeds while maintaining the exclusion of forgiven amounts from gross income. This created a substantial tax windfall: business owners could exclude PPP from income while still deducting the wages and other expenses that PPP funded. Also in late December 2020, Congress enacted a second draw of PPP to be disbursed in 2021, intended to be more targeted to smaller firms that experienced at least a 25 percent decline in receipts in at least one quarter of 2020.

The economic literature finds that the PPP played a role in stabilizing employment during the pandemic, though at a high cost per job saved. Autor et al. (2022*b*) estimate that PPP eligibility increased employment by 2–5 percent at its peak, retaining 3.6 million jobs mid-2020, with effects fading to 1.4 million jobs by year-end. The cost per job-year saved ranged between \$170,000 and \$258,000, significantly exceeding the median worker’s annual earnings. Despite favoritism in loan allocation by banks (Bartik et al. (2020)), the speed of PPP disbursement was critical in preventing further economic damage, as delays would have exacerbated job losses. Granja et al. (2023) find even more modest employment effects at the county level, with implied costs per job saved surpassing \$175,000. Splinter et al. (Forthcoming) finds that the program retained 3.8–4.2 million job-years, with a gross cost of \$133,000–\$146,000 per job-year, and estimates that about 55 percent of PPP forgiveness went to workers. Examining the initial roll out of the program, Doniger and Kay (2021) find that delays in loan disbursements led to persistent employment losses.

While this literature focuses specifically on employment effects—and less so on wages and profits needed to understand incidence—the apparent high cost per job saved suggests PPP provided a substantial benefit to firm owners.

3 Methods

To construct our data, we first merge loan-level microdata from the Small Business Administration for all PPP loans disbursed in 2020 including the Employer Identification Number

(EIN) or Social Security Number (SSN) of the borrower to de-identified IRS income tax records of the business entities that received those loans including sole proprietors filing Schedule C with their individual returns; C- and S- Corporations filing forms 1120 and 1120s; and partnerships filing form 1065, as well as to employment tax returns (including Forms 941, (the Employer’s Quarterly Federal Tax Return), 943 or 944), which are filed by employers to report employee tax withholding. The 941 is filed by almost all U.S. employers, including nonprofit employers and employers in U.S. Territories, who do not file income tax returns. Form 941 was the primary recommended documentation to supply to PPP lenders to substantiate payroll. Additionally, in cases where there is no match, we also check for any EIN application (Form SS-4) and, in case a taxpayer identification number was misclassified, we also attempted to merge loan EINs to SSNs in the tax data and vice-versa.

Our sample covers the universe of approximately 5.2 million PPP loans disbursed in 2020, totaling \$529 billion. We match these loans to IRS business tax records by EIN or SSN. We successfully match about 84 percent of PPP dollars to business tax filings. The unmatched 16 percent includes approximately 2.5 percent flowing to nonprofit organizations (which have no shareholders), 8 percent to entities that filed employment tax returns (Form 941) but whose income tax returns could not be identified (potentially including businesses in U.S. territories), 1.4 percent to partnerships and corporations whose owners could not be identified, 1 percent to corporate subsidiaries that could not be linked to parent firms, and 3.5 percent that could not be classified. In our distributional analysis, we gross up aggregate amounts within each income category and business type to reflect the full population, an approach that is reasonable to the extent that the unmatched entities are distributed similarly to matched entities within each category.

Second, we link the information on recipient businesses to the owners of those businesses. For sole proprietorships, this link is direct, because these owners file their business returns with their individual returns. For pass-through business owners (S-corporation and partnership owners), we attribute ownership based on ownership shares (inputted for partnerships

from distribution shares) on Schedule K-1 issued by businesses to their owners, and for C-corporations we use the information of Schedule G, which lists ownership shares of major owners, along with distribution shares among those issuing Form 1099-DIV. Because the match rate of the merge between SBA data and tax records is central to the discussion of ineligible payments, we describe characteristics of the matched sample below.

4 Analysis

4.1 Matching PPP Disbursements to Eligible Firms

Matching PPP loans to tax records of recipient firms provides useful information about the share of loans that flowed to eligible firms, and the correspondence of each PPP loan to the characteristics of the business. In particular, we decompose the total amount of recorded PPP loans in 2020 (\$529 billion) into those amounts that did not appear to flow to actual businesses and, among the remainder, the amounts that flowed to businesses in excess of statutory limits.

The first step in this analysis proceeds directly from the match between the universe of SBA records and the universe of business records. Table 1 shows that about \$16.4 billion of the \$529 billion total PPP disbursed in 2020 (4 percent) flowed to individuals or entities that have no corresponding IRS records—firms that simply did not exist. The next row in Table 1 shows that an additional \$13.8 billion accrued to entities that do appear in IRS records, but have no payroll or qualifying expenses that would qualify them for any PPP. In particular, these entities have no employee compensation in 2019 or early 2020 (either on income tax or employment tax returns), no 2019 net earnings subject to self-employment tax among general partners reported on Form 1065 K-1 (nor on Form 1040 Schedule SE), and no 2019 positive self-employment income reported on Form 1040 Schedule C or F. In short, about 6 percent of PPP funds in 2020 seem to have accrued to fictitious businesses or those with no payroll expenses.

Table 1: PPP Disbursements by Eligibility Criteria (\$ billions)

Total PPP Disbursements (2020)	528.9
Eligibility Classification	
Is Recipient a Business?	
Not a qualifying business entity	16.4
No payroll or qualifying expenses	13.8
<i>Total non-business recipients</i>	<i>30.2</i>
Does Business Meet PPP Eligibility Rules?	
Employment Exceeds Qualifying Threshold	9.1
Not in operation in February 2020	8.9
<i>Total non-qualifying payments</i>	<i>18.0</i>
Was PPP Amount Disbursed Correct?	
PPP Payments In Excess of Qualifying Payroll	43.8
Were PPP Disbursements Used for Qualifying Expenses?	
Amount of PPP Disbursements Not Used for Qualifying Expenses	5.6
Total Ineligible PPP Paid	97.6
As a % of total PPP payments	18.9%

Source: Small Business Administration and Treasury.

Note: All amounts in \$ billions. “Potentially ineligible” per criteria described in Section 4; misclassification and measurement error may remain. “No matching business tax record” includes entities with no corresponding IRS filing. “No payroll or qualifying expenses” includes entities with no employee compensation, self-employment income, or partner earnings in IRS records.

Our estimates are broadly consistent with other assessments of PPP program integrity. The SBA Office of Inspector General has estimated over \$200 billion in potentially fraudulent disbursements across PPP and related pandemic programs (Small Business Administration Office of Inspector General, 2024), and the Government Accountability Office has documented widespread fraud schemes and screening failures (Government Accountability Office, 2022). We interpret these data as evidence of one form of PPP fraud. For instance, when we tabulate rates of PPP payments to potentially ineligible entities by lender (Appendix Table A1), we observe high rates of such payments at certain “fintech” financial institutions (which is consistent with other estimates of fraud Griffin, Kruger and Mahajan (2023)). In fact, executives at the two lenders with the highest measured rate of ineligible payments

in the data were convicted for PPP fraud. Similarly, we examine how the rate of ineligible payments varies with the timing of application (Appendix Figure A.1). While ineligible payments started out low (around 5 percent), they increased to over 30 percent of payments by the close of the program. We suspect that most initial transactions occurred between financial institutions and their well-known, preexisting clients, but that individuals seeking to defraud the government learned gradually about the program integrity vulnerabilities and entered later.

However, we also find significant amounts of ineligible payments arising from other sources. Of the remainder of PPP funds, a significant share appear to have been paid to actual businesses with active operations as of February 15, 2020, but which were ineligible to receive PPP based on the parameters described in PPP legislation.

In particular, \$9.1 billion was paid to businesses whose level of employment from employment tax returns (or for a small share of industries, revenue from income tax returns) exceeded the maximum thresholds in the legislation and whose assets and net income exceeded the alternative eligibility computation. Another \$8.9 billion was disbursed to businesses that were not in operation in February 2020 (as measured from Form SS-4 and conservatively from employment and income tax returns), and thus should have been ineligible.

In theory, the ineligible disbursements described above should have been prevented by lender due diligence. Under the CARES Act and related SBA guidance, lenders were required to verify, at a minimum, that the applicant business was legitimate, had been in operation on or before February 15, 2020, met SBA size criteria, and had verifiable payroll expenses aligning with the requested loan amount. This involved verifying that the payroll documentation provided by the borrower matched claimed figures and that no duplicate PPP loans were sought.

A different form of ineligible payment arises when eligible firms receive PPP loans in excess of the amount specified by law or when the PPP loans are not used for qualifying expenses. We find that about \$44 billion in PPP was paid out in amounts in excess of those

specified based on payrolls expenses specified by Treasury/SBA regulations, which we recreate for every firm in our sample.³ The maximum PPP loan amount was primarily calculated by taking 2.5 times a borrower’s average monthly payroll costs (typically based on 2019 figures or the trailing 12 months), subject to a cap of \$10 million. “Payroll costs” included wages, salaries, tips, certain employee benefits, and state and local taxes assessed on compensation, but excluded annualized compensation over \$100,000 for any individual employee. The \$45 billion is the amount by which PPP disbursements exceeded those limits. Hence, the largest source of apparent non-compliance appears to be excessive PPP disbursements at otherwise eligible firms. Lenders were permitted to let potential recipients calculate this amount themselves and substantiate with documentation that they could self-certify to be accurate.

Finally, another important PPP requirement was that the PPP funds actually be used to support spending on qualifying expenses, like payroll and certain other costs. In short, PPP recipients had to use PPP funds or retain their employees and support business operations. However, some firms received PPP and nevertheless fired their workers and/or stopped making qualifying payments. The last row in Table 1 suggests about \$5 billion of PPP was disbursed to qualifying firms but is not eligible for forgiveness due to too little spending on employee payroll (as measured from Q2 and Q3 quarterly payroll).

In sum, of the \$529 billion in first round PPP, about \$100 billion were ineligible payments, or about 19 percent of the total. Our analysis suggests that information available at the time when PPP was implemented could have been used to identify and prevent ineligible payments. For example, at the time of PPP disbursement, firms had already filed Form 941 from the fourth quarter of 2019 and they soon would file it for the first quarter of 2020 by April 30th. Likewise, the characteristics of participating lenders and the duration of the

³For firms where we observe employment tax returns but no income tax returns, we impute employee benefits. For firms where we observe income tax returns but not employment tax returns, we impute cash compensation in excess of \$100,000, income excluded from Medicare wages, mainly employee health insurance premiums, and the potential benefit of using the most recent 12 months instead of 2019 as the payroll lookback period. When the figures do not agree between employment tax returns and income tax returns, we use the higher of the two.

program also appear to have affected ineligible payments.

4.2 How Did PPP Affect Firm Profits?

To understand how PPP affected firm profits, we start with a descriptive analysis of the revenues, expenses, and profits of firms that received PPP over the period from 2019 to 2020. This analysis steps through the calculation of pre-tax, pre-PPP profits to post-tax, post-PPP profits in each of the years.

Table 2 describes the key elements of the average revenue, cost, and profits in 2019 and 2020 of the 3.9 million corporations, partnerships and sole proprietorships that received PPP in 2020.

In 2019, on average, these firms earned receipts of about \$2.3 million, paid expenses of about \$2.2 million, of which about \$499,000 was wages, and earned average net income of \$101,600. On this income, in 2019, firms paid about \$29,000 in federal income taxes, resulting in average after-tax income of about \$73,000.

In 2020, total receipts declined modestly, by about 2.6 percent, but so did non-labor costs; hence net income declined to \$95,000. Tax payments fell slightly, hence after-tax, but pre-PPP income fell to \$67,000.

However, in 2020, these firms received an average PPP amount of \$103,000. Recall that PPP disbursements were excluded from income, while expenses paid by PPP (e.g. to workers) remained tax deductible (and taxable to workers). From a measurement perspective, this makes business receipts and net income appear artificially low in the affected years. In reality, the after-tax, after-PPP income of businesses was much higher in 2020, and increased much more, from \$73,000 in 2019 to \$171,000—more than 2.3 times greater.

To illustrate how profits changed across the distribution of all PPP recipient firms, Figure 1 presents kernel density estimates of net income in 2019, 2020 excluding PPP, and 2020 including PPP. The effect of the pandemic is evident in the leftward shift in the distribution of income between 2019 and 2020 (excluding PPP), and the spike in net income around zero.

Table 2: How Did PPP Affect Pre- and Post-Tax Profits of Businesses?

Corporations, Partnerships, and Sole Proprietors			
Income Item	2019 (1)	2020 (2)	% Change (3)
Total Receipts	2,258,926	2,200,785	-2.6%
Less			
Payroll and Compensation	499,126	502,634	0.7%
Other Expenses	1,658,163	1,603,399	-3.3%
Total Expenses	2,157,290	2,106,034	-2.4%
Equals			
Pre-tax, pre-PPP Income	101,637	94,751	-6.8%
Less			
Taxes	28,714	27,567	-4.0%
Equals			
After-Tax, Pre-PPP Income	72,923	67,184	-7.9%
Plus			
PPP		103,386	
Equals			
Pre-Tax Net, plus PPP income	101,637	198,137	94.9%
After-Tax, plus PPP Income	72,923	170,570	133.9%
N	3,876,763	3,876,763	

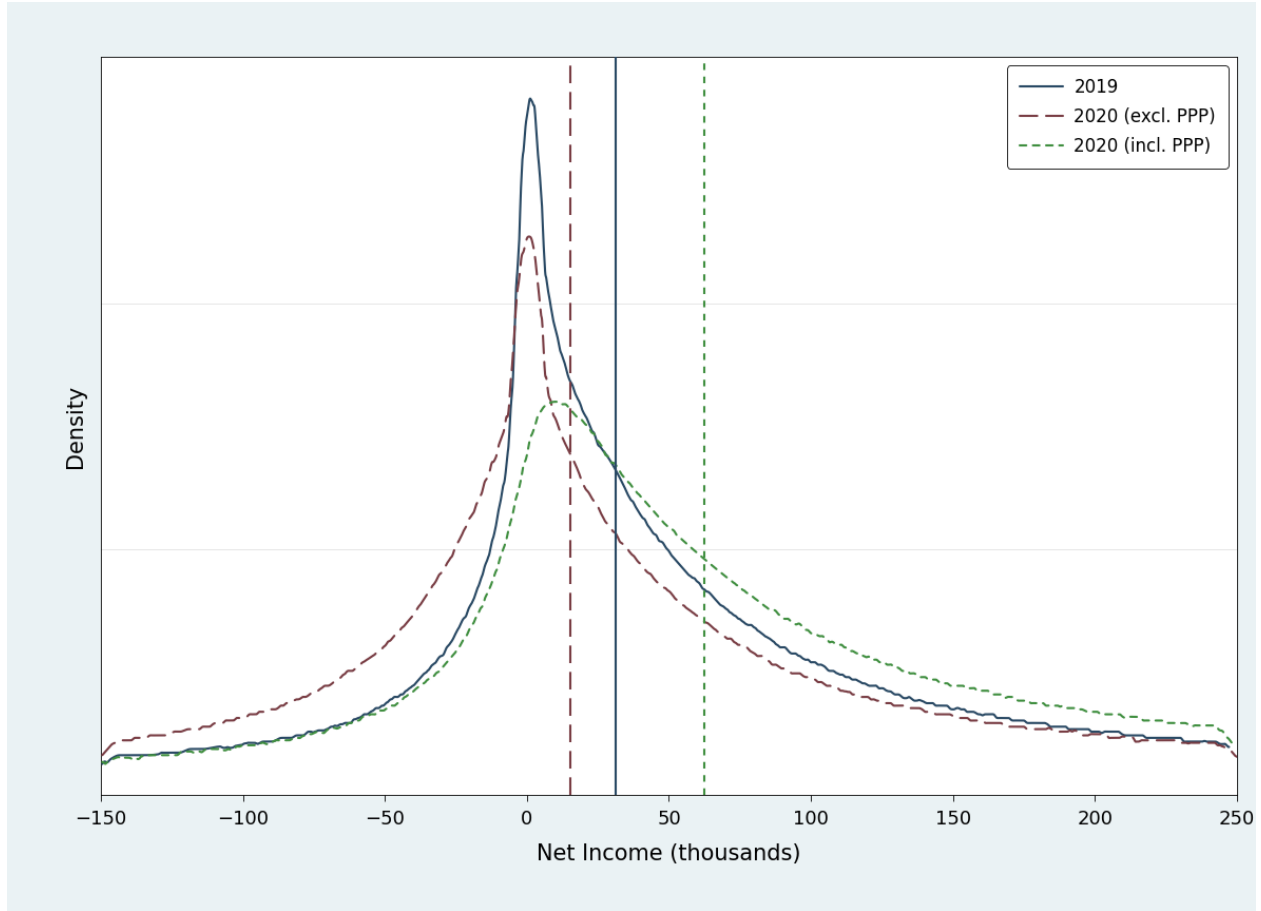
Note: All amounts are mean dollars per recipient firm in nominal dollars. “Taxes” are federal income taxes attributable to business income: entity-level for C-corporations and owner-level for pass-through entities, allocated based on ownership shares from IRS return data. Source: IRS and Small Business Administration.

Including PPP, however, the entire distribution shifts substantially to the right, reflecting the large average PPP amounts relative to net income shown in Table 2.

A similar pattern is apparent in receipts; the typical change in business receipts between 2019 and 2020 was small and the change in receipts at most firms was close to zero (whether unweighted or weighted by 2019 receipts). Including PPP, however, most firms gross receipts were higher than in 2019.

The fact that the entire distribution of income and receipts is shifted rightwards suggests

Figure 1: Distribution of Net Income, 2019–2020



Note: Kernel density estimates of net business income for PPP recipient firms. Blue: 2019 net income. Red: 2020 net income excluding PPP. Green: 2020 net income including PPP. Vertical lines indicate the median net income for each distribution. Source: IRS and Small Business Administration.

that PPP payments benefited firms regardless of their economic situation during the pandemic. Indeed, 40 percent of firms had higher receipts in 2020 than in the prior year, and these firms received about 41 percent of PPP. (Note that even in a normal year many firms experience falling receipts; in 2019, only 62 percent had higher receipts than in 2018). The 43 percent of firms which had higher employment in 2020 than in the prior year received 47 percent of PPP; and 60 percent of (ex-PPP) profitable firms received 62 percent of PPP. (See Appendix Table B1.)

These tabulations suggest, first, that the impact of the pandemic was surprisingly modest relative to baseline volatility in year-to-year changes in income, employment, and receipts.

Second, the amount of PPP received by firms was not particularly targeted at firms experiencing declines in revenues, employment, or profits.

Likewise, in other analysis, we find that PPP takeup rates and average PPP amounts received was relatively uniform across industries—whereas the impact of the pandemic on revenues, costs, and profits varied significantly across industries. More surprisingly, when we compare the changes in business activity among firms that received PPP to firms that appear eligible for PPP but did not take up PPP, their economic experiences during the pandemic were relatively similar. For instance, the decline in receipts among non-recipients was -1 percent versus -3 percent for recipients; the decline in expenses was -1 percent vs -2 percent, and the change in profits before PPP was -1 percent versus -6 percent. After including PPP, however, recipient firms’ profits increased 134 percent. In short, PPP had very large effects on recipients even compared to those who appeared eligible but who did not take up the program. Finally, our analysis suggests that among eligible firms, larger firms received more PPP (largely because they had larger payrolls), but among these larger firms, the increase in after-PPP income was also greater, suggesting that the value of PPP more than compensated for reductions in receipts or net income. (See Appendix B.)

4.3 Second Round

In late 2020, Congress enacted a second round of PPP intended to be more targeted to smaller firms and those more affected by pandemic disruptions (see Appendix C for eligibility details). Table 3 presents the income elements in 2019, 2020, and 2021 for firms that received the second round.

The second draw indeed reached firms with larger 2020 revenue declines: these 995,000 businesses experienced an average 20 percent decline in receipts and an 89 percent decline in pre-tax net income in 2020, compared to average increases of 5 percent in receipts and 27 percent in pre-tax profits among firms that received only the first round. However, by the time the second round was actually distributed, these firms’ receipts and payroll had already

Table 3: The 2021 Second Draw PPP

Corporations, Partnerships, and Sole Proprietors					
Income Item	2019	2020	2021	%Change 2019-2020	%Change 2019-2021
	(1)	(2)	(3)	(4)	(5)
Total Receipts	2,423,236	1,939,883	2,417,158	-19.9%	-0.3%
Less					
Payroll and Compensation	565,369	498,900	560,167	-11.8%	-0.9%
Other Expenses	1,762,262	1,430,292	1,754,633	-18.8%	-0.4%
Total Expenses	2,327,631	1,929,192	2,314,800	-17.1%	-0.6%
Equals					
Pre-tax, pre-PPP Income	95,604	10,691	102,357	-88.8%	7.1%
Less					
Taxes	27,524	6,946	29,845	-74.8%	8.4%
Equals					
After-Tax, Pre-PPP Income	68,080	3,745	72,512	-94.5%	6.5%
Plus					
PPP		117,128	117,149		
Equals					
Pre-Tax Net, plus PPP in-	95,604	127,819	219,506	33.7%	129.6%
come					
After-Tax, plus PPP In-	68,080	120,872	189,661	77.5%	178.6%
come					
N	994,858	994,858	994,858		

Note: All amounts are mean dollars per recipient firm in nominal dollars. “Taxes” are federal income taxes attributable to business income: entity-level for C-corporations and owner-level for pass-through entities, allocated based on ownership shares from IRS return data. Source: IRS and Small Business Administration.

recovered to pre-pandemic levels. As a result, after-tax, PPP-inclusive profits were nearly three times 2019 levels in 2021.

5 Incidence

The preceding analysis documented the *mechanical* effects of PPP on firm profits—that is, the accounting impact of adding tax-exempt PPP receipts to firms’ income statements. We now turn to the question of *behavioral* incidence: to what extent did the benefits of PPP accrue to firm owners versus employees? Because we do not observe a credible counterfactual for what would have happened absent PPP at the firm level, we do not pursue a causal identification strategy. Instead, we provide several bounding exercises that yield a consistent range of estimates under transparent assumptions.

The observed data are difficult to reconcile with majority-worker incidence under any plausible counterfactual. After-tax business profits rose 134 percent while worker compensation increased 0.7 percent. For workers to have captured the majority of PPP benefits, counterfactual wages would need to have fallen by more than the entirety of PPP disbursements—implying that absent the program, firms would have cut compensation by far more than the modest revenue declines they actually experienced. We formalize this intuition with several bounding exercises.

Looking back, the evidence above suggests the pandemic’s impact on firm revenues and profits was temporary and modest, and a large share of firms saw their business activity increase. For businesses that experienced little interruption, workers’ marginal products would have been unaffected and firms would have retained workers absent PPP, making PPP a windfall to owners.

We attempt to bound the incidence to owners using several approaches. For these exercises, we ignore general equilibrium effects and focus on the incentives introduced by the program to retain payroll.

Program rules. During the covered period, only 75 percent of PPP funds needed to be allocated toward payroll; the remaining 25 percent accrued to owners. Additionally, the exclusion of PPP from taxable income benefited owners: retaining payroll was fully subsidized even without the tax exclusion, because PPP-funded wage expenses remained

deductible. Since the tax exclusion saved about \$145 billion (on \$430 billion of PPP), combining this with the non-payroll allowance implies at least 44 percent of the total PPP “value” accrued to owners mechanically.

Firms with increased receipts. About 40 percent of PPP recipient firms had higher gross receipts in 2020 than in 2019 (Appendix Table B1). These firms would have been unlikely to lay off workers absent PPP, since their marginal products did not decline. Combining PPP at these firms with the tax and program rules above, we estimate that at least 68 percent of benefits accrued to owners.

Firms with increased profits. We can also consider the share of PPP that increased after-tax profits relative to 2019. Assuming profits would have weakly declined absent PPP given the pandemic’s disruption, we estimate a lower bound of 67 percent accruing to owners.

External estimates. Econometric studies suggest PPP reduced job loss by 2 to 5 percent at the pandemic’s peak (Autor et al., 2022b). At the average firm, 5 percent of payroll is about \$29,000—roughly 22 percent of the average PPP amount—which still implies the vast majority of benefits flowed to owners. This likely understates owner incidence because retained workers had positive marginal products but imposed no marginal cost on firms, since their compensation was paid by PPP.

In sum, multiple approaches suggest that between two-thirds and three-quarters of PPP benefits accrued to firm owners. Even if no specific assumption is unassailable, the convergence of results across approaches supports this conclusion. Our range of 67–75 percent to owners contrasts with Splinter et al. (Forthcoming)’s estimate that 55 percent went to workers, which relies primarily on estimated employment effects rather than observed firm-level profit and compensation data.

6 The Distribution of PPP and after-tax PPP Among Business Owners

Having established that two-thirds to three-quarters of PPP benefits accrued to owners, we now describe how those benefits were *distributed* across the income distribution. This is an accounting allocation exercise: we assign PPP amounts to individual owners based on observed ownership shares and rank owners by income. To produce this analysis, we match the PPP recipient businesses to the individual owners of these businesses. For Schedule C filers, whose business returns are attached to their individual tax returns, this matching is direct, and the PPP dollars are associated directly to the filer. Schedule C filers represent about 9 percent of PPP dollars in our sample.

For other businesses, we apportion PPP to owners based on the share of the business they own. About 15 percent of PPP flowed to C-corporations. For C-corporations, we use information from Schedule G which provides information on the ownership shares of major shareholders.

For owners of partnerships and S-corporations, we use Schedule K-1 information returns to link the recipient businesses to their owners, and allocate the PPP income among owners in proportion to their share of ownership. About 60 percent of PPP dollars accrued to such pass-through business.

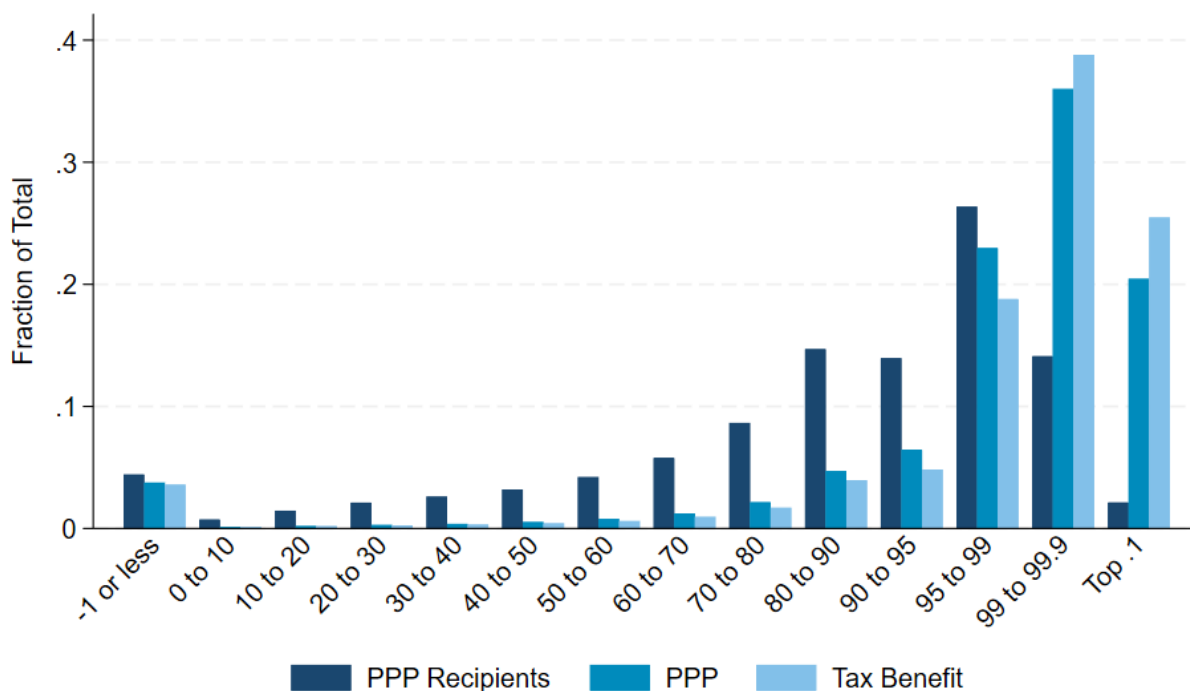
We were unable to match about 16 percent of dollars. About 2.5 percent appear to be nonprofit businesses, who have no shareholders. About 1.4 percent are partnerships and corporations whose owners could not be identified (perhaps because they were owned indirectly by other partnerships or by foreign owners). About 8 percent appeared to flow to entities that filed a Form 941 Employer's Quarterly Federal Tax Return, but whose business tax return could not be identified (these may include businesses in U.S. territories that file employment tax returns but not federal income tax returns). One percent are corporate subsidiaries that don't file on their own but could not be linked to their parent firm. The

remaining 3.5 percent could not be identified. In our analysis, we gross up the aggregate amounts in each income category accruing to each business type to reflect total distributions. This assumes that unmatched dollars are distributed across income categories in the same proportion as matched dollars within each business type. If unmatched dollars were disproportionately concentrated among lower-income owners, our estimates would overstate top-end concentration; however, the unmatched share is modest (16 percent).

Figure 2 summarizes the results of this analysis. The chart categorizes the recipients of PPP funds by their place in the 2020 income distribution (ranked by pre-tax income including PPP) and shows the share of recipients, the share of pre-tax, pre-PPP income earned by those recipients, the share of total PPP received by each group, and the share of the tax savings associated with the exclusion of PPP from tax.

Figure 2: Who Benefited from PPP?

Receipt, Income, PPP, and Tax Benefit by Income Percentile.



Note: Percentiles are based on 2020 total income including PPP, which reflects the distribution of realized resources. The share of PPP recipients, pre-tax pre-PPP income, PPP amounts, and tax savings from the PPP exclusion are shown for each income group. Source: IRS and Small Business Administration.

The figure shows that PPP-recipient business owners are disproportionately concentrated at higher income levels. For example, of the roughly 7 million individual business owner PPP recipients we identify, about 16 percent are in the top 1 percent of the income distribution based on their 2020 income including PPP. (Ranked by 2019 or 2020 income excluding PPP, the shares are 14 and 13 percent, respectively.) Likewise, more than half of all PPP recipient owners were in the top 10 percent of the income distribution based on their incomes in 2019 or 2020. Because business income is so concentrated, among PPP recipients, the top 1 percent business owners earned about 68 percent of all pre-tax income in 2020 and the top 10 percent about 93 percent of all income.

(Based on 2020 total IRS-reported income, individuals needed to earn about \$151,000 to be in the top 10 percent, \$556,000 to be in the top 1 percent, and \$2.6 million to be in the top 0.1 percent.)

Figure 2 also describes the share of PPP flowing to each income group. About 56 percent of PPP funds (\$243 billion) accrued to owners in the top 1 percent and 86 percent to owners in the top 10 percent (or \$370 billion) (when recipients are ranked by their PPP-inclusive income; ignoring PPP about 37 percent flowed to the top 1 percent based on 2019 or 2020 income, and about 70 percent to the top 10 percent).

Finally, we also assess how much the unusual tax treatment of PPP benefited individuals at different income levels. Because PPP was excluded from income, it avoided taxation at the individual income tax rate that would otherwise have applied to their business income. Hence the tax benefit is likely to be even more concentrated in high-income groups than PPP itself because of the progressive marginal tax structure, which applies higher rates to higher-earning individuals.

According to this analysis, the beneficial tax treatment of PPP saved recipients almost \$145 billion in taxes in 2020—about a third of the \$430 billion in PPP they received. About two thirds of the benefit (about \$93 billion dollars) accrued to the top 1 percent and almost 90 percent to the top 10 percent (\$127 billion).

As a result, PPP clearly had a large effect on the disposable income and economic well-being of these business owners. In 2019 and 2020, in the aggregate, PPP recipient business owners reported total after-tax income (excluding PPP) of \$1.8 trillion and \$1.9 trillion, respectively. In 2020, including PPP they earned \$2.4 trillion—about 30 percent more than in 2019. (And that is income after paying workers and incurring other business expenses.)

PPP-recipient business owners in the top 1 percent of the income distribution took home \$1.2 trillion after tax in 2019, but \$1.7 trillion after tax, after PPP in 2020—about 42 percent more.

To put these magnitudes in perspective, the Tax Cuts and Jobs Act of 2017—the largest tax legislation in three decades—increased the after-tax income of top-1-percent households by an estimated 3.4 percent in 2018.⁴ PPP’s distributional effect on high-income business owners was thus roughly twelve times larger than TCJA’s, albeit temporary rather than permanent.

7 Conclusion

The Paycheck Protection Program (PPP) played an important role in stabilizing employment and supporting businesses at the onset of the pandemic, but its design and implementation revealed key areas for improvement.

Summary of findings

This paper documents three central findings about the PPP using the universe of PPP loans linked to IRS tax records. First, about \$100 billion (19 percent) of 2020 PPP disbursements were potentially ineligible—flowing to fictitious entities, ineligible businesses, or in amounts exceeding statutory limits—and much of this was identifiable from government records available at the time of disbursement. Second, among eligible recipients, after-tax business profits

⁴Tax Policy Center estimate of the conference agreement, December 2017.

rose 134 percent while worker compensation increased just 0.7 percent; bounding exercises suggest that two-thirds to three-quarters of program benefits accrued to business owners rather than workers. Third, PPP benefits were concentrated among the wealthiest business owners: over half of PPP dollars flowed to owners in the top 1 percent of the income distribution, boosting their after-tax income by 42 percent in a single year. The exclusion of forgiven PPP from taxable income saved owners an estimated \$145 billion, with about two-thirds of that benefit flowing to the top 1 percent, magnifying this concentration.

Why the program design produced these outcomes

Several features of PPP’s design contributed to these results. Loan amounts were calibrated to 2.5 months of payroll, which for many firms exceeded the actual disruption caused by the pandemic—particularly as economic activity recovered in the second half of 2020. The PPP Flexibility Act further reduced the effective payroll maintenance requirement, making forgiveness conditions largely nonbinding. Broad eligibility criteria, combined with reliance on borrower self-certification rather than verification against existing government data, allowed substantial ineligible disbursements. Finally, the late-2020 Congressional reversal of the IRS’s initial position on deductibility created an unintended double tax benefit that magnified the windfall to owners.

Policy implications

These findings suggest several lessons for the design of future emergency business support programs.

First, screening using existing administrative data—such as IRS payroll filings (Form 941), employer identification applications (Form SS-4), and business tax returns—could have substantially reduced ineligible payments with minimal delay to legitimate disbursements. Our analysis shows that much of the information needed to flag fictitious or ineligible recipients was available in government databases before funds were disbursed.

Second, targeting mechanisms such as revenue-decline thresholds could improve the efficiency of emergency programs. The second round of PPP attempted this by requiring a 25 percent decline in quarterly receipts, and indeed reached firms more affected by the pandemic. However, by the time funds were disbursed, many of these firms had already recovered, illustrating the challenge of timely targeting.

Third, the tax treatment of forgiven PPP created a substantial and regressive windfall. Either including forgiven PPP in gross income or disallowing the deduction of PPP-funded expenses would have reduced the program's fiscal cost by approximately \$145 billion while maintaining the core employment-stabilization objective. Including forgiven amounts in income is the more straightforward approach, as it preserves standard tax accounting while still allowing full loan forgiveness.

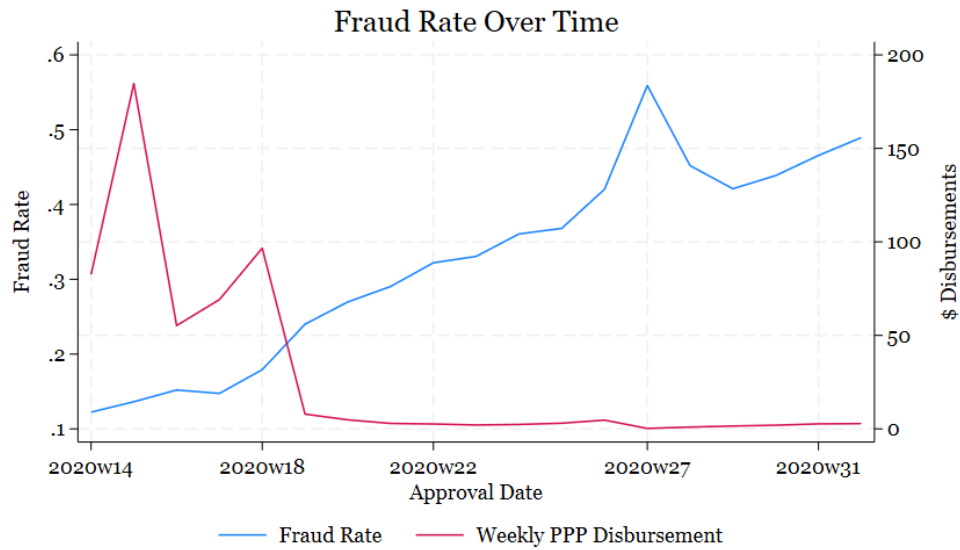
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A Appendix

A. Potentially Ineligible Payments and Lender Analysis

Figure A.1: Share of Disbursements Flagged as Potentially Ineligible, by Approval Date



Source: Small Business Administration and author's calculations.

Appendix Table A1: Potentially Ineligible Disbursements by Lender

Largest Lenders

Note: “Ineligible” denotes loans to entities with no matching business tax record, no payroll, employment exceeding qualifying thresholds, or not in operation in February 2020. “Improper” denotes loans to otherwise eligible firms where the disbursed amount exceeded statutory limits or funds were not used for qualifying expenses. PPP \$ and amounts in \$ billions.

Type of Lender	PPP Loans	PPP \$	Ineligible (\$)	Improper (\$)	% Ineligible	% Improper	Total
National Bank	343,582	25.5	1.8	2.4	7%	10%	17%
National Bank	280,157	29.5	2.3	3.3	8%	11%	19%
Fintech	268,683	8.1	2.3	0.9	29%	11%	40%
National Bank	194,344	10.7	0.8	1.1	7%	10%	17%
Fintech	126,258	2.4	1.6	0.2	64%	10%	74%
Regional Bank	108,326	7.6	0.7	0.8	9%	10%	19%
National Bank	108,210	5.3	0.8	0.9	16%	17%	33%
National Bank	85,960	8.6	0.4	1.0	5%	11%	16%
National Bank	82,042	13.2	0.6	1.5	4%	11%	16%
Fintech	80,589	0.9	0.1	0.1	16%	10%	27%
Fintech	76,150	2.2	0.4	0.2	18%	10%	29%
National Bank	73,923	13.1	0.5	1.4	4%	11%	15%
Specialized/Fintech Bank	66,591	3.1	0.9	0.5	29%	15%	44%
Regional Bank	50,977	5.0	0.3	0.7	5%	13%	18%
Regional Bank	47,826	7.0	0.4	0.6	5%	9%	14%
Regional Bank	47,184	4.7	0.5	0.5	11%	10%	21%
Regional Bank	43,168	8.2	0.5	0.9	6%	10%	16%
Regional Bank	40,064	5.5	0.3	0.6	6%	11%	17%
Regional Bank	38,484	6.6	0.4	0.7	6%	11%	16%
Regional Bank	34,649	6.8	0.3	0.6	4%	8%	12%
Regional Bank	32,764	4.2	0.2	0.3	6%	7%	13%
National Bank	30,923	3.4	0.3	0.4	8%	12%	20%
Regional Bank	27,546	1.2	0.2	0.1	13%	5%	17%
Fintech	23,925	0.6	0.4	0.1	62%	14%	75%
Regional Bank	23,353	3.2	0.1	0.2	3%	5%	8%

Source: Small Business Administration.

Appendix Table A1 (Continued): Potentially Ineligible Disbursements by Lender

Lenders with Highest Rates of Potentially Ineligible Disbursements (among lenders with 5,000+ loans)							
Type of Lender	PPP Loans	PPP \$	Ineligible (\$)	Improper (\$)	% Ineligible	% Improper	Total
Specialized / Fintech	23,925	0.6	0.4	0.1	62%	14%	75%
Fintech	126,258	2.4	1.6	0.2	64%	10%	74%
Fintech	14,271	0.3	0.2	0.0	44%	10%	54%
Specialized / Fintech	66,591	3.1	0.9	0.5	29%	15%	44%
Specialized / Fintech	268,683	8.1	2.3	0.9	29%	11%	40%
Regional	108,210	5.3	0.8	0.9	16%	17%	33%
Specialized / Fintech	76,150	2.2	0.4	0.2	18%	10%	29%
Fintech	80,589	0.9	0.1	0.1	16%	10%	27%
Specialized	11,672	1.3	0.1	0.1	11%	11%	22%
Specialized / Fintech	19,928	0.7	0.1	0.1	13%	9%	22%
Regional	47,184	4.7	0.5	0.5	11%	10%	21%
National	30,923	3.4	0.3	0.4	8%	12%	20%
National	108,326	7.6	0.7	0.8	9%	10%	19%
National	280,157	29.5	2.3	3.3	8%	11%	19%
Regional	18,868	3.1	0.2	0.4	7%	12%	19%
Regional	50,977	5.0	0.3	0.7	5%	13%	18%
Regional	16,649	2.0	0.1	0.2	6%	12%	18%
Regional	27,546	1.2	0.2	0.1	13%	5%	17%
National	194,344	10.7	0.8	1.1	7%	10%	17%
Regional	40,064	5.5	0.3	0.6	6%	11%	17%
National	343,582	25.5	1.8	2.4	7%	10%	17%
National	85,960	8.6	0.4	1.0	5%	11%	16%
Regional	38,484	6.6	0.4	0.7	6%	11%	16%
Regional	43,168	8.2	0.5	0.9	6%	10%	16%
National	22,759	3.3	0.2	0.3	7%	9%	16%

Source: Small Business Administration.

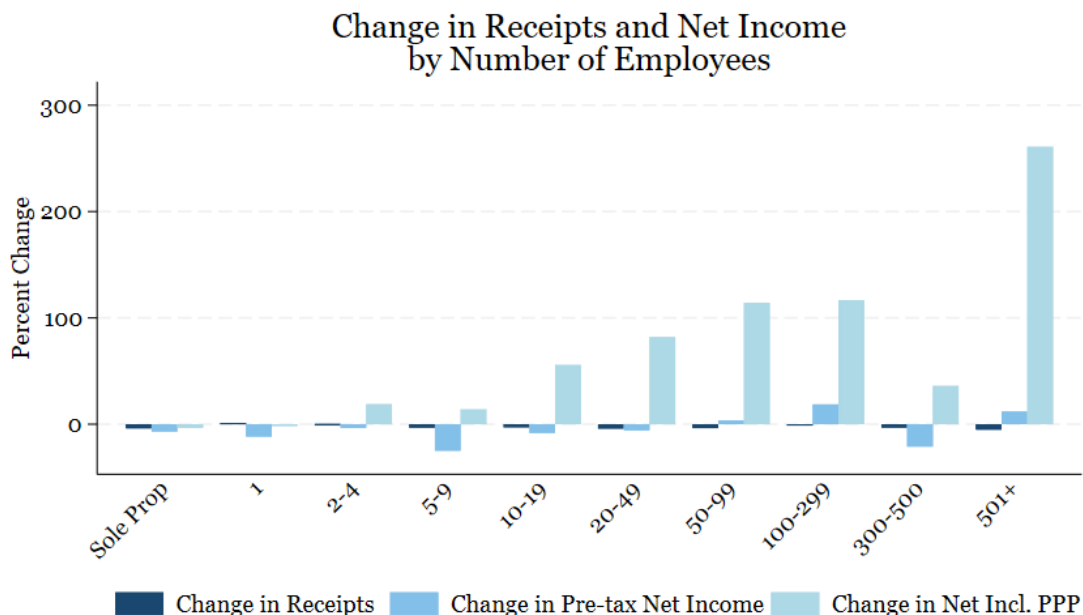
Appendix Table B1: Distribution of PPP by Firm Characteristics

	All PPP Re- cipients	Higher Re- ceipts	Re- employment	Higher Em- ployment	Higher Pre- PPP Profits	Positive Prof- its	Higher After- Tax, After PPP Profits
Number of Firms	2,803,335	1,116,870		1,212,271	1,138,321	1,676,294	1,943,465
Percent of PPP Recip- ient Firms	100%	40%		43%	41%	60%	69%
Average PPP	131,608	135,810		142,645	141,962	136,336	153,174
% of all \$PPP	100%	41%		47%	44%	62%	81%
Baseline: Character- istics of all Eligible Firms	Number of Firms		CY Receipts > LY Re- ceipts		CY Profit > LY Profit	CY Profit > 0	
Pre-pandemic: All 2019 eligible firms (2018 versus 2019)	4,588,180		2,825,661		2,956,806	3,218,205	
Percent	100%		62%		64%	70%	
Pandemic: All 2020 eligible firms (2019 versus 2020)	4,703,898		2,339,315		2,651,002	2,932,620	
Percent	100%		50%		56%	62%	
Pandemic: 2020 eli- gible firms that did not receive PPP (2019 versus 2020)	1,900,563		1,222,445		1,512,681	1,256,326	
Percent	100%		64%		80%	66%	

B. Firm Size Analysis

Figure B1 presents the change in receipts and net income by firm size (measured by number of employees). While receipts declined modestly across all firm sizes, the change in net income including PPP was substantially positive, particularly for larger firms. This pattern suggests that PPP benefits were proportionally larger for firms with more employees, consistent with the program’s design of tying loan amounts to payroll.

Figure B.1: Change in Receipts and Net Income by Firm Size



Source: IRS and Small Business Administration. Note: Firm size measured by number of employees.

C. PPP Program Rules

Effective payroll maintenance threshold. PPP loans were sized at 2.5 times average monthly payroll. Under the revised rules, 60 percent of the loan must be spent on payroll over 24 weeks (approximately 6 months). For a firm with monthly payroll of P , the loan equals $2.5P$ and the payroll requirement is $0.6 \times 2.5P = 1.5P$ over 6 months, or $0.25P$ per month—about 25 percent of prior monthly payroll. Accounting for rounding and the precise covered-period length, the effective threshold is approximately 27 percent. Few firms would have been constrained by this low threshold.

Safe harbors. The Flexibility Act provided safe harbors for borrowers that could document an inability to rehire former employees or return to pre-pandemic activity levels due to COVID-19 compliance requirements. Combined with the extended covered period, these provisions made forgiveness conditions largely nonbinding for most recipients.

Appendix Table C1: PPP Program Rules

Provision	Original CARES Act	PPP Flexibility Act (June 2020)
Covered period	8 weeks	24 weeks
Payroll share for forgiveness	75% of loan proceeds	60% of loan proceeds
Compensation maintenance	75% of Q1 2020 levels	Safe harbors for decreases
Employment restoration deadline	June 30, 2020	December 31, 2020
Eligible expenses (non-payroll)	Mortgage interest, rent, utilities	Same
Maximum loan	2.5× avg. monthly payroll (\$10M cap)	Same
Employee compensation cap	\$100,000 annualized per employee	Same
Eligible employers	≤500 employees (or SBA size standard)	Same

Second-draw eligibility. The second round of PPP, enacted in late December 2020, was available to businesses with 300 or fewer employees that had experienced at least a 25 percent decline in gross receipts in any quarter of 2020 relative to the same quarter in 2019. Maximum loan amounts were generally calculated as 2.5 times average monthly payroll (or 3.5 times for accommodation and food services businesses), subject to a \$2 million cap.